JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR COLLEGE OF ENGINEERING (AUTONOMOUS) ANANTHAPURAMU– 515002 (A.P) INDIA
Department of Computer Science and Engineering
II B.Tech I Semester (R23)

Subject Code	Title of the Subject	L	T	P	C
23A35105	**DISCRETE MATHEMATICS & GRAPH THEORY**	**3**	**0**	**0**	**3**

Course Objectives:

Introduce the concepts of mathematical logic and gain knowledge in sets, relations and functions and Solve problems using counting techniques and combinatory and to introduce generating functions and recurrence relations. Use Graph Theory for solving real world problems.

Course Outcomes:

After successful completion of this course, the students should be able to:

COs	Statements	Blooms level
CO1	Apply mathematical logic to solve problems.	L2, L3
CO2	Understand the concepts and perform the operations related to sets, relations and functions. Gain the conceptual background needed and identify structures of algebraic nature.	L3, L5
CO3	Apply basic counting techniques to solve combinatorial problems.	L3
CO4	Formulate problems and solve recurrence relations.	L2, L3
CO5	Apply Graph Theory in solving computer science problems	L3, L5

CourseArticulationMatrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	1
CO2	3	2	2	2	-	-	-	-	-	-	-	1
CO3	3	2	2	1	-	-	-	-	-	-	-	1
CO4	2	2	2	1	-	-	-	-	-	-	-	1
CO5	3	3	2	1	-	-	-	-	-	-	-	1

1-Slightly, 2-Moderately, 3-Substantially

UNIT – I: sets and propositions

Introduction, combinations of sets, finite and infinite sets, uncountably infinite sets, principle of inclusion and exclusion, mathematical induction, propositions, logical connectiviteies, conditional and biconditionals,well formed formulas, tautologies, logical equaliences, Normal forms, theory of inference for statement calculus, predicate calculus ,the statement function, variable and quantifiers.

UNIT II Relations and functions:

Introduction, a relation model for data bases, properities of binary relations, closure of relations, warshels algorithm, equilence relations and partitions, partial ordering relations and lattices, functions, composition of functions, invetiable function, recursive functions.

UNIT – III: Permutations, combinations and Recurrence relations

Introduction, the rules of sum and product, permutations and combinations

Recurrence relations and recursive algorithms: Introduction, recurrence relations, linear recurrence relations with constant coefficients, homogenous solutions, particular solutions, total solutions, solution by the method of generating functions.

UNIT – IV : Groups and Rings:

Introduction, groups, sub groups, normal subgroups, generators and equitation of powers, cosets, codes and group codes, isomorphisms and automorphisms, homomorphisms and, rings ,integral domains, and fields, ring homomorphisms, polynomial rings .

Boolean algebra: lattices and algebraic systems, principle of duality, basic properties of algebraic systems defined by lattices,

UNIT – V: Graph Theory

Basic terminology, multigraphs and weighted graphs, diagraphs and relations, reprentation of graphs, operations on graphs, paths and circuits, graph Traversals, shortest paths in weighted graphs, eulerianpaths and circuits, planar graphs

Textbooks:

1. C L Liu| D P Mohapatra, Elements of Discrete Mathematics A computer Oriented Approach, Fourth Edition, McGraw-Hill Education.
2. J.P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, 2002

ReferenceBooks:

1. Kenneth H. Rosen, Discrete Mathematics and its Applications with Combinatorics and Graph Theory, 7th Edition, McGraw Hill Education (India) Private Limited.
2. Joe L. Mott, Abraham Kandel and Theodore P. Baker, Discrete Mathematics for Computer Scientists & Mathematicians, 2nd Edition, Pearson Education
3. Graph Theory with Applications to Engineering and Computer Science by Narsingh Deo.

Online Learning Resources:

1. http://www.cs.yale.edu/homes/aspnes/classes/202/notes.pdf